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Effect of Feed Combination Variation on BSF Larvae Weight and Extraction Method on BSF Larvae Oil Yield in Biodiesel Production

Adrianto Prihartantyo ^{1*}, Nina Roberta ², Aga Nugraha ³, Astiti Aditia⁴,
Fitriyanti Aruan ⁵, Naomi Harianja⁶, and Tama Sihombing⁷

^{1, 4, 5, 6, 7} Department of Bioprocess Engineering, Faculty of Biotechnology, Institut Teknologi Del, Sitoluama, Toba, 22381, Indonesia

² Department of Chemical Engineering, Institut Teknologi Sumatera, Jl. Terusan Ryacudu, Way Huwi, Kec. Jati Agung, Lampung Selatan 35365, Indonesia

³ Department of Metallurgy Engineering, Faculty of Biotechnology, Institut Teknologi Del, Sitoluama, Toba, 22381, Indonesia

*E-mail: adrianto.prihartantyo@del.ac.id

Abstract. Black soldier fly (BSF) larvae have a high capacity to decompose organic waste and contain a high fat content, making them a potential raw material for biodiesel. The weight of BSF larvae is influenced by optimal feed conditions, which include waste that has decomposed for 2 days, cut into 5-8 cm pieces, and has a moisture content of 70-80% w/v. High-fat feeds such as skipjack tuna (23.85% w/w), avocado (15% w/w), and tofu (13.72% w/w) are provided in a 1:1 combination. The oil from BSF larvae is extracted using soxhlet and maceration methods. The research results show that the feed combination can increase the weight of BSF larvae, with the avocado:tofu combination yielding the highest weight. The Soxhlet method produced the highest oil yield of 61.33% w/w and showed the best selectivity for producing lauric, myristic, and oleic acids, which are important for biodiesel. The biomass from BSF larvae oil produced in this study meets the biodiesel standard SNI 7128:2015

1. Introduction

The demand for biodiesel in Indonesia continues to increase along with economic growth and population. Currently, biodiesel production in Indonesia only contributes about 13% of the total national fuel needs, and this figure is still far from the government's target of 23% by 2025 [2]. On the other hand, Indonesia is also facing the problem of continuously accumulating organic waste. According to data from the Ministry of Environment and Forestry in 2021, the amount of organic waste is estimated to reach 11.7 million tons. One solution that can address both of these issues is the utilization of Black Soldier Fly (BSF) larvae as an alternative raw material for biodiesel. BSF larvae can be fed with organic waste, resulting in a biomass rich in fat [8]. The fat content in BSF larvae is about 40-45% w/w [11]. The fat content in BSF larvae can be increased by providing feed that is high in fat [3]. The research conducted by [1] used feed from high-fat organic waste, namely skipjack tuna, avocado, and tofu. The fat content in each of the feed formulations was 23.1% w/w; 15% w/w; and 13.72% w/w. From the feeding, the fat content of the harvested BSF larvae was 47.73% w/w; 43.51% w/w; and 41.31% w/w, respectively [1].

The feeding conditions in the study [1] were conducted in an uncontrolled manner, therefore in this study, the feeding conditions are provided in a controlled manner. The controlled feed condition is that the feed must undergo decomposition for 2 days to create a pungent smell that is favored by BSF larvae [13]. Next, the feed must be chopped into sizes of $\pm$ 5-8 cm, because BSF larvae do not have a functional mouth for the eating process [13]. Finally, the feed must have an optimal moisture content 70-80% w/v [4], as the moisture in the feed is necessary to dissolve and extract the feed, so that it can be well absorbed by the larvae.



The increase in fat content in BSF larvae is marked by the increase in the weight of BSF larvae. The weight of BSF larvae can be increased by combining feed that is high in fat under controlled conditions. In this study, BSF larvae will be given a 1:1 feed combination of skipjack tuna, tofu, and avocado. After 21 days of cultivation, BSF larvae can be harvested and oil can be extracted from their bodies using solid-liquid chemical extraction methods, namely soxhlet extraction and maceration. In this study, the effect of extraction methods on the amount of BSF larva oil and the selectivity of extraction towards the percentage of lauric, myristic, and oleic fatty acids in BSF larva oil was observed. Lauric, myristic, and oleic acids contained in BSF larva oil have the potential to be raw materials for biodiesel. Lauric and myristic acids play a role in producing biodiesel with low viscosity and low pour point due to their short carbon chains [15]. Oleic acid can produce biodiesel with a low flash point, thereby increasing the oxidation stability of biodiesel [15]. And in this research, biodiesel production was carried out from BSF larva oil a local food ingredient and one of the pillars of global food security. One of the palm sugar derivatives with the highest economic value is palm crystal sugar, which is used as a raw material for a large proportion of food and beverages and is commonly consumed all over the world. In 2023, the global consumption of palm sugar will amount to approximately 177.33 million metric tons (MMT), and an annual growth of 0.8% is forecast [1]. However, the global production of palm sugar in 2023 was only 80.4 MMT. According to national palm sugar statistics, domestic consumption is

2. Experimental Procedures

2.1 Making BSF Larvae Feed

Skipjack tuna, avocado, and tofu are first washed with water to remove any dirt that may be present on the surface of the organic waste, before being fermented for 2 days. After that, the organic waste is chopped using a knife until it measures approximately 5-8 cm. The skipjack tuna is chopped into all parts of the body, including the head and bones. Similarly, with avocados, where the entire avocado is used, including the avocado seed. The chopped organic waste is checked for moisture content in the range of 70-80% w/v using a moisture meter.

2.2 Feeding BSF Larvae

6-day-old BSF larvae, approximately 2000 of them obtained from BSF larvae breeders located in Medan, are weighed first before being cultivated. Once their weight is measured, they can be cultivated for 9 days by providing feed to the BSF larvae. Cultivation begins at 6 days old because BSF larvae are ready to digest organic waste at 6 days after hatching from BSF eggs. The control feed is given 1.5 kg each, while the 1:1 combination feed is given 0.75 kg of each type of feed to the BSF larvae cultivation container.

2.3 Harvesting and Weighing BSF Larvae

BSF larvae that are 21 days old are then harvested, where the harvesting is done by separating the BSF larvae from the waste in the cultivation container. The harvested BSF larvae are weighed using an analytical balance.

2.4 Washing BSF Larvae

BSF larvae are washed using hot water at a temperature of 70°C. The purpose of this washing is to clean off any residual dirt adhering to the BSF larvae's bodies while also killing them.

2.5 Drying and Grinding BSF Larvae

The drying of BSF larvae is carried out using an oven at a temperature of 60°C for 24 hours, while the grinding of BSF larvae into powder is done using a blend

2.6 Extraction of BSF Larva Oil and Solvent Purification

In the soxhlet extraction process, 60 grams of BSF larva powder and 128.57 ml of n-hexane solvent were used at a temperature of 69°C. The soxhlet extraction process was stopped when the solvent color in the soxhlet flask became clear again. In the maceration process, 60 grams of BSF larva powder and 300 ml of n-hexane solvent were used. The maceration process was also accompanied by stirring using a magnetic stirrer, with a rotation speed of 300 rpm for 20 minutes every 4 hours. The purification of the solvent was carried out using a rotary evaporator at a temperature of 40°C.

2.7 Fatty Acid Content Test

The fatty acid content test was conducted by sending BSF larva oil samples to the Universitas Airlangga laboratory, and the testing was performed using a Gas Chromatography device (GC).

2.8 Biodiesel Production

Biodiesel production is carried out through a transesterification process using 30 ml of BSF larva oil, 52.82 ml of methanol, and 1% KOH catalyst. After that, a washing process is conducted using 50% v/v aquades. The final process is heating the biodiesel using a hot plate at 110°C, and stirring with a magnetic stirrer at a rotation speed of 200 rpm for 30 minutes.

3. Results and Discussion

3.1 Feed Combination Increases the Percentage of BSF Larvae Weight

Based on Figure 1, it can be seen that the feed combination has been proven to increase the percentage of BSF larvae weight.

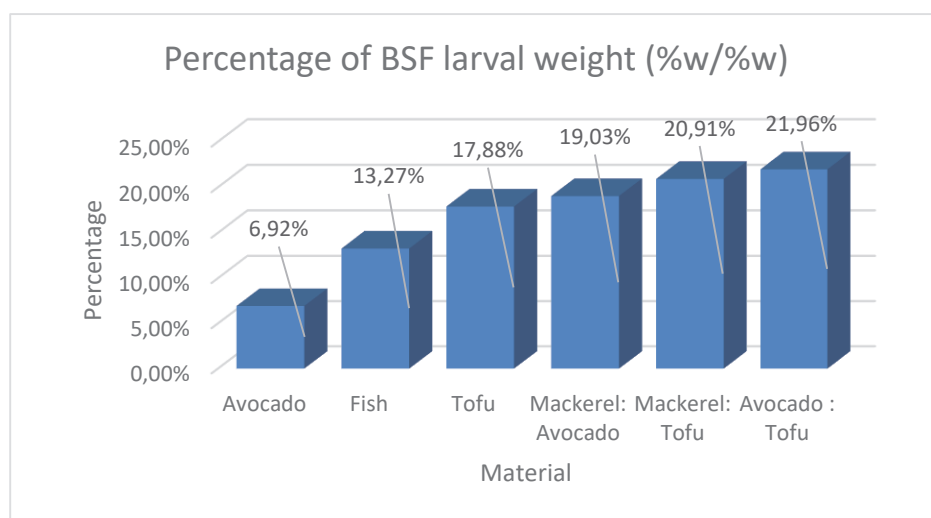


Figure 1. Graph of the percentage increase in BSF larva weight.

This occurs due to the higher fat content in the feed combination. The higher fat content provides a very high energy source for the BSF larvae's body, as fat is high in calories [12]. The calorie content in fat is 9 calories per gram, which is higher than the calorie content in protein and carbohydrates, which is only 4 calories per gram [12]. This high calorie content allows BSF larvae to obtain more energy from each gram of feed consumed, thereby increasing their body weight.

Additionally, based on Figure 1, it can be seen that the combination of avocado and tofu feed provides the highest percentage of BSF larva weight. This can happen because in the type of feed, there are parts that cannot be consumed by BSF larvae, specifically the hard components of the feed. The hard feed cannot be consumed by BSF larvae because, primarily, BSF larvae do not have a functional mouth like teeth to chew

the feed [5]. Tofu can be fully consumed by BSF larvae because tofu has a feed component that is 100% soft, allowing it to be completely consumed.

In skipjack tuna, it cannot be fully consumed by BSF larvae because skipjack tuna has a soft feed component of only 50-60% w/w, where this component is the fish flesh and its internal organs [9]. In avocado, it cannot be fully consumed by BSF larvae because avocado has a soft feed component of only 70-80% w/w, where this component is the fruit flesh [10]. From the three types of feed, it can be determined which feed is the most consumed. The order of the most consumed feed is tofu, avocado, skipjack tuna.

3.2 Soxhlet Extraction Produces the Most BSF Larva Oil

Based on Figure 2, it can be seen that Soxhlet extraction produces the most BSF larva oil compared to maceration.

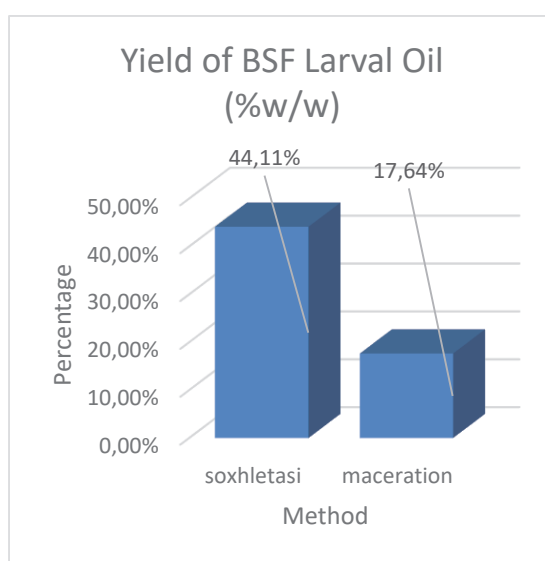


Figure 2. Yield of BSF larvae oil from various extraction methods

This can occur because the soxhlet extraction process uses high temperatures (boiling point of the solvent), compared to the maceration process which is conducted at room temperature. The high temperature in the soxhlet extraction process can increase the solubility of the oil in the solvent, allowing more oil to be extracted [9]. This is also supported by the statement [6] that high soxhlet extraction temperatures can cause greater solvent movement, due to the increased kinetic energy between molecules. Additionally, high temperatures can also accelerate the diffusion process, allowing more oil from the sample to be extracted. The situation is different with the maceration method conducted at room temperature, which is less effective in increasing the solubility of oil in the solvent, resulting in less oil being extracted [9]. This is also supported by the statement [6] that the maceration process conducted at room temperature causes less movement of the solvent, as this occurs due to the lower increase in kinetic energy between molecules.

3.3 Soxhlet Extraction Produces the Best Selectivity

Based on Figure 3, it can be seen that Soxhlet extraction produces the best fatty acid selectivity compared to maceration.

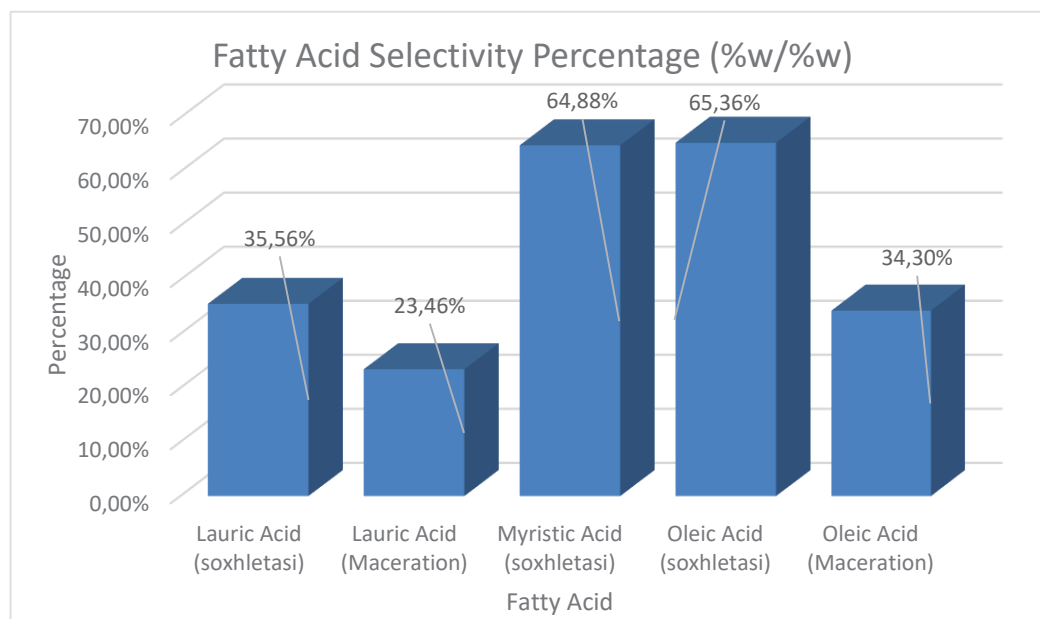


Figure 3. Fatty acid selectivity of BSF larva oil from various extraction methods

This is because in soxhlet extraction, it works at high temperatures (the boiling point of the solvent), thereby increasing the solubility of fatty acids in the solvent. Solvent molecules have higher kinetic energy, making them more effective at dissolving a large amount of fatty acids from the sample matrix [14]. This is different from maceration, which works at room temperature, resulting in lower solubility of fatty acids in the solvent. This condition causes solvent molecules to have less kinetic energy, making them less effective at dissolving a large amount of fatty acids [14].

Additionally, the high temperature during soxhlet extraction also causes the breaking of van der Waals bonds, which are the bonds that bind fatty acids in the sample matrix [7]. By breaking these bonds, the solvent can more easily extract fatty acids. In maceration, it occurs at room temperature, making it difficult to break the van der Waals bonds in the sample matrix. Therefore, only a small amount of fatty acids were extracted using the maceration method [7].

3.4 Biodiesel from BSF Larva Oil

In this study, the biodiesel characteristics have been tested according to the SNI 7128:2015 standard. The results of the biodiesel characteristic tests from BSF larva oil are shown in Table 3.1.

Table 3.1. Characteristics of biodiesel from BSF larva oil

No	Parameter	Unit	Standart	Test Result
1	Density at 20°C	Kg/m ³	850-890	868
2	Kinematic viscosity at 40°C	mm ² /s (cSt)	2.3-6.0	4.26
3	The number of the beast	-	Min 51	51
4	Acid number	Mg-KOH/g	Maks 0.5	0.32
5	Iodine number	%-massa	Maks 115	104.81

Based on Table 3.1, it can be concluded that the test results of the biodiesel characteristics from BSF larva oil meet the SNI 7128:2015 standards, thus it can be stated that BSF larva oil can be used as a raw material for biodiesel production.

4. Conclusion

The combination of feeds can increase the percentage of BSF larval weight compared to each individual feed. The feed combination that yields the highest percentage of BSF larval weight is avocado: tofu. The extraction method that yields the most oil from BSF larvae is Soxhlet extraction, with a yield of 44.11% w/w. The extraction method with the best selectivity is Soxhlet extraction. BSF larva oil can be used as a raw material for biodiesel.

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